Vario USB Bench Programme

Comprehensive Engineering Report — Phases 1–5 and the t008 Quiet-House Static Validation

Dual-barometer + IMU variometer benchmark on an ESP32-S3 rig: hardware, datasets, causal replay methodology, 29-filter comparison, audio behaviour, fault analysis, real-IMU fusion, and cross-generation static noise characterization.

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Evidence labels: [M] measured · [S] simulated · [D] derived proxy · [H] hypothesis · [U] unverified

Executive summary

This report consolidates a five-phase bench programme that took a variometer sensor rig from first wiring to a validated real-IMU fusion pipeline with a fully reproducible filter benchmark. Principal measured results:

• The two GY-63 MS5611 clones share a per-sample pressure noise floor of 2.6–2.9 Pa (≈0.21–0.24 m altitude) at OSR2048/90 Hz, ~2–3× the datasheet typical; a quiet-house static (t008) proves this floor is sensor-intrinsic, while the office capture (t007) shows 7× more slow drift (9.2 m vs 1.3 m range) caused by room-pressure draft [M].

• A baro-only constant-velocity Kalman filter at an exploratory q=10 with measured r reaches a rest velocity noise floor of 0.10–0.11 m/s RMS on quiet statics across three firmware generations [M].

• Of 29 standalone filters benchmarked causally, robust/adaptive Kalman variants (S1/S5) dominate the rest-noise/transient-rejection trade; S5 matches S1 at ~1/3 CPU [M]. The fastest exit-from-lift is S4 (median release proxy 1.43 s) [D].

• The S2 dual-timescale filter froze permanently after landing transients — root-caused to a plausibility pre-clean that replaced deviating samples with its own prediction; the saturation fix (never replace, always move toward the sample) eliminates the trap [M].

• Real-IMU fusion (Phase 5): with an exact attitude warm-start and loss-of-track reinitialization, the fused estimator equals baro-only at rest and on elevator motion, and degrades gracefully on drop/toss — the limiter is 6-DOF attitude quality under tumble (6.5–10.3 m/s² gravity residual), not the Kalman filter [M].

• A portable C99 core reproduces the Python reference estimator within 2.0e-08 m/s (double build) and cross-compiles for ESP32-S3 [M/S].

• No final estimator is selected or frozen in this report, per programme instruction; q=10 is an exploratory operating point only.

1. System overview

1.1 Hardware as-built

The bench rig is an ESP32-S3 with two GY-63 modules (MS5611 barometric clones) on separate I²C buses and one GY-521 (MPU6050 IMU). Clone #1 has CSB pulled low and answers at 0x76 on bus A (GPIO4/5) together with the MPU6050 at 0x69 (AD0 tied high, WHO_AM_I=0x68, genuine die). Clone #2 — the unit under test — answers at 0x77 on bus B (GPIO6/7). A third barometer position (bit-banged bus C, GPIO15/16) exists on the schematic but is not populated. PROM[0] reads 0x0878 (bus A) and 0x0884 (bus B). The rig is mounted X-up: at rest the dominant specific-force component is ax ≈ +1 g, so no analysis may assume az is the gravity axis [M].

Table 1. *Bench rig hardware as-built [M].*

Module	Bus	Address	Role / notes
GY-63 #1 (MS5611 clone)	A (GPIO4/5)	0x76	CSB pulled low; sid 0
GY-521 (MPU6050)	A (GPIO4/5)	0x69	genuine; ±8 g / ±500 dps; DLPF 44 Hz
GY-63 #2 (MS5611 clone)	B (GPIO6/7)	0x77	unit under test; sid 1
GY-63 #3 (optional)	C (GPIO15/16)	—	not populated

1.2 Firmware generations

Table 2. *Firmware generations. All speak serial protocol v1 over USB-CDC at 921600 baud: boot legend, per-device health check, PROM dump, raw frames only — no on-MCU filtering.*

Generation	Version	Stream	Role
t001	0.1.3	baro only	bring-up, noise floor (T-001)
t002	0.2.2	baro + 200 Hz IMU	IMU path validation (T-002/T-004)
t003	0.3.1	dual baro lockstep 45 Hz	dual-sensor study (T-003)
t005	0.5.3	dual baro 50 Hz	OSR4096 motion set (T-005)
t006	0.6.0	dual baro measured 90 Hz	OSR2048 high-rate set
t007/t008	0.7.0	dual baro 90 Hz + IMU 200 Hz	real-IMU fusion dataset (OSR2048)

Every frame carries a sequence number and a physical MCU timestamp (esp_timer_get_time); the offline pipeline derives all sampling intervals from these timestamps, never from USB receipt time. A JSON sidecar per capture carries firmware config and PROM words; offline temperature/pressure compensation is impossible without it.

1.3 Capture and analysis pipeline

Captures are taken with `tools/capture.py` (COM5 @ 921600). The benchmark replay engine feeds identical chronological (sample, physical-timestamp) pairs to every filter; corrupt frames are band-filtered and counted, sequence gaps are preserved in diagnostics, and the filters experience true variable dt. The two barometers are replayed as independent streams unless an explicitly defined fusion algorithm is being evaluated. Operational lessons (serial-monitor port locks, CDC non-reset behaviour, meta-on-attach timing, label reuse) are recorded in docs/HANDOVER.md.

2. Datasets

22 recordings are committed under analysis/data/ (CSV + JSON sidecar), spanning statics, elevator rides, step drops, toss-and-catch and controlled hand motion across firmware generations t001–t008. Session types come from operator labels — naming, not independent evidence. The machine-generated quality verdict for every run is in filter_benchmark/reports/data_quality_report.md; the summary is reproducible via scripts/02_validate_data.py.

Table 3. *Recording inventory by session class (counts of captures).*

Dataset	Statics	Elevator	Drop	Toss	Controlled	IMU
t001–t002	3	—	—	—	—	t002 only
t003 dual	1	—	—	—	—	—
t004	—	2	1	—	—	yes
t005 OSR4096	1	1	1	1	1	—
t006 OSR2048	1	—	1	—	1	—
t007 OSR2048+IMU	1	1	2	1	1	yes
t008 quiet static	1	—	—	—	—	yes

Data-quality highlights [M]: the t008 quiet static is clean (one 1-frame sequence gap). Motion sessions from t007 lose 3.5–9 % of sample time to 3–12 simultaneous freezes of both barometers and the IMU (0.32–0.66 s each) — the coincidence across all three streams pins the cause to whole-loop USB/serial backpressure, not I2C. These freezes double as accidental blackout-coast test intervals. Drop/toss impacts saturate the ±8 g accelerometer for <1 % of IMU samples (noted, expected at impacts). One t004 elevator capture lacks its sidecar and is excluded from all compensated analysis.

2.1 Reference channels

No capture contains GPS, a reference-vario output stream, or ground-truth event tags. All event timings reported below are descriptive proxies against derived references (1-s Gaussian-smoothed barometric derivative with strong-motion gating), never physical latency. The spec-mandated B03 reference-vario baseline is therefore Unavailable in every recording and is reported as such [U].

3. Benchmark methodology

3.1 Causal replay and evidence labels

Filters are evaluated by causal replay on physical timestamps with whole-session tuning/validation separation (9 training + 5 validation sessions in Phases 2–4; multiple sids per session are not independent samples). Every number carries an evidence label: [M] measured on bench captures, [S] simulated on seeded synthetic streams, [D] derived proxy against a derived reference, [H] hypothesis/design default, [U] unverified. Per-filter tuning was fit on training sessions only and reported against held-out validation sessions; every tested configuration is preserved in results/ (nothing discarded).

3.2 Estimator model

The core estimator is a timestamp-driven asynchronous Kalman filter over state x = [h, v] (constant-velocity model) with IMU propagation extending the state usage in fusion mode:

$$F = [[1,\ dt],\ [0,\ 1]],\quad Q = q\,[[dt^3/3,\ dt^2/2],\ [dt^2/2,\ dt]],\quad H = [[1,\ 0]]$$

with process spectral density q and measurement variance r. In fusion mode (Phase 5) each timestamped 3-axis accel sample propagates the state through gravity-compensated Earth-vertical acceleration and each baro sample corrects it; accel bias is estimated only under a rest gate (|v| < 0.5 m/s and |a_comp| < 1 m/s², clamped to ±2 m/s²). Innovation gating (3.5σ) plus a loss-of-track reinitializer (8 consecutive >5σ innovations ⇒ reacquire from the measurement) protect against transients. Altitude comes from the MS5611 second-order temperature-compensated conversion and the ISA troposphere relation h = 44330·(1–(p/p■)^0.1903) with p■ from the first ~40 s of a session.

3.3 Audio evaluation

Audio behaviour is measured through a separate state machine (src/audio.py) so filter and audio defects are independently attributable. The common policy across Phases 2–4 is lift-on +0.20 m/s, lift-off +0.05 m/s; Phase 4 adds a generic fault veto (5-of-5 streak detector) and a max_on valve. False-ON rates are per-hour on statics; exit-from-lift is first-class (an ON-at-event-end with no OFF within 8 s is censored and counted, never dropped); continuous beeping under sustained faults is measured as ON-fraction with and without the veto.

4. Phase 1–2: baselines and the 29-filter standalone benchmark

Phase 1 delivered the dataset inventory and the machine-checked quality report. Phase 2's mandate was correctness and comparability, not selection: all baselines and all standalone filters (Groups A smoothing, B classical, C trackers, D Kalman variants, E adaptive) were run on identical streams with default and tuned configurations (unit tests 11/11; 29 filters registered; 667 individual-filter rows).

Table 4. *Baselines B00–B03 at defaults, session medians [M]; proxies [D].*

Baseline	Static rest RMS [m/s]	Motion RMS [m/s]	Onset proxy [s]	Release proxy [s]	False latches
B00 raw diff-quota	9.140	11.630	−2.905	7.940	1
B01 regression slope	0.159	1.277	−1.965	4.765	7
B02 project Kalman [M]	0.096	0.563	−0.100	4.885	1
B03 reference vario	n/a — Unavailable in all recordings [U]				

Representative standalone results at default configurations (session medians; full 26-row table in reports/phase2_demonstration.md): alpha-beta trackers C01/C02 are quietest at rest (0.026/0.021 m/s) but blind at onset (proxy 1.2–1.4 s); the CV Kalman D01 reaches 0.101 m/s rest with onset proxy −0.31 s; robust/adaptive variants D04/D06 and E06 improve motion handling (0.58–0.92 m/s motion RMS); windowed smoothers (SMA/Median/Hampel) trade 1.3–2.4 m/s motion RMS for 0.25–0.49 m/s rest noise; fixed high-order low-passes (Butterworth/Bessel) ring badly at rest (2.5–3.4 m/s) — the classic pressure-smoothing trap the background research warns about [M][D].

4.1 Per-filter tuning and held-out validation

Each filter was tuned on training sessions (rank-based multi-objective score: static RMS, motion RMS, onset proxy, false latches) and validated held-out. Tuned configurations improved motion RMS for most filters (e.g. A02 EMA 0.85→0.58 m/s, C02 ABG 0.63→0.22 m/s) while validation onsets stayed consistent — no filter was tuned to the validation split [M].

4.2 Rate comparison and its correction

A Phase 2 table appeared to show 1.4–1.6× higher rest noise at 90 Hz than 45 Hz for windowed smoothers. Phase 3 audited this: trailing windows are parameterised in seconds, for which rate cancels exactly ($v(t) = (z(t) - z(t-W))/W$); synthetic controls show no rate effect, and decimating the t006 static 90→45 Hz changes SMA rest RMS by only 1.6 %. The apparent effect was firmware noise colour (OSR4096 vs OSR2048; increment lag-1 correlation −0.28 vs −0.47) — a session-pair confounder, not a rate effect [M]. The Phase 2 numbers stand; the interpretation is corrected here.

5. Phase 3: stacked candidates and audio behaviour

Three stacked architectures were frozen before any held-out run (configs in config/phase3.json with per-parameter provenance): S1_RobustAdaptiveKF (Hampel pre-clean + adaptive-gain CV Kalman), S2_DualTimescale (fast EMA + slow gated KF with persistence exit), S3_AccelerationAwareKF (3-state [h,v,a] KF with fault detection). Reference baselines B02/C03/D01 ran with frozen configs for comparability.

Table 5. *Phase 3 estimator + audio results [M] (train statics; val proxies [D]). *S2's 0.00 s applies only to the 42 % of events where it exited at all — on the other 57 % its persistence latch held ON past the window (the freeze trap, §6.1).*

Filter	Rest RMS [m/s]	Onset proxy [s]	Beep-off proxy [s]	False ON/hr	ON outside events
S1_RobustAdaptiveKF	0.045	0.14	2.06	60	1.0
S2_DualTimescale	0.081	–0.27	0.00*	120	2.0
S3_AccelerationAwareKF	0.334	–3.90	0.27	13260	221
B02_project_kalman	0.092	–0.02	1.76	480	8.0
C03_CV_tracker	0.048	1.29	0.96	0	0.0
D01_Kalman_CV	0.096	–2.35	0.92	120	2.0

Seeded synthetic fault injection (spike / burst / gap / reversal; recorded-failure threshold 1.0 m/s) showed S1/S2 rejecting all four transients while C03 and D01 recorded failures on the burst scenario (2.99/3.00 m/s) and baro-only S3 failed all four [S]. CPU cost (host Python, relative is the portable quantity): S1 42.9 µs/update (2.95× D01) due to its sliding Hampel window; S2 16.6 µs (1.14×); C03 6.6 µs (0.45×) [M].

5.1 Recorded failures (never hidden)

S3 chatters audio at ~13 000 ON/hour on statics — its baro-only acceleration state excites the velocity output — and reaches 16.2 m/s post-transient on corrupt frames; S1 has one marginal 1.04 m/s excursion; D01/C03 record excursions on the same transients. All ten failure rows are preserved in results/08_audio_failures.csv [M].

6. Phase 4: corrected metrics, safety mechanisms, C core

6.1 The S2 freeze trap — root cause and fix-by-design [M]

Corrected per-event accounting (873 events) exposed 31 censored beep-offs on S2 — audio that never exited after an event, 17 lift / 13 sink / 1 weak. Root cause: S2's plausibility pre-clean computed each incoming sample's innovation against the fast branch's own prediction and, beyond max_dev, replaced the sample with the prediction. Under the sustained deceleration of a landing, the real signal departs from the prediction faster than the prediction follows; once the gap exceeded max_dev, every real sample was discarded (median 2017 rejected samples on censored events), the filter consumed only its own output, and (h,v) froze at a wrong value — median tail v = 0.449 m/s with audio pinned ON to end of capture.

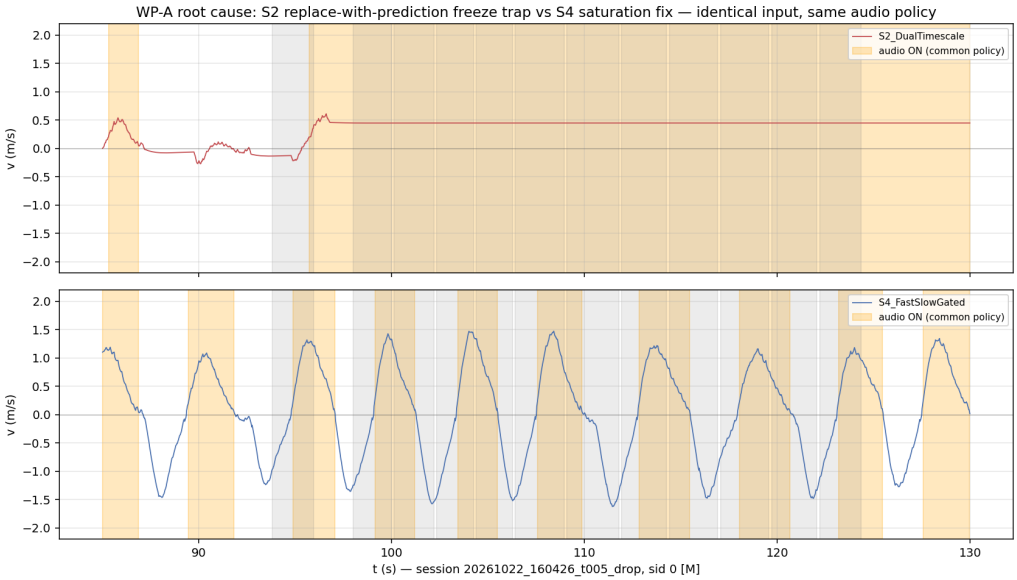


Figure 1. *The identical drop-session stream through frozen S2 (top) and the saturation-fixed S4 (bottom). S2 freezes after the landing transient and never exits; S4 recovers and exits each event normally [M].*

Fix applied in all subsequent filters: saturate, never replace — zc = h_pred + sign(dev)·max_dev. The filter always moves toward the real signal at max_dev per sample and can never become self-referential; S4_FastSlowGated has zero censored events on identical input. S2/S3 are frozen untouched as defect references so Phase 3 stays reproducible.

6.2 Retained candidates (Phase 4)

Table 6. *Phase 4 estimator comparison, frozen configs, common audio policy [M]; proxies [D]. †S1's two policy-level censored events are removed by the tuned+safety policy. S1's missed-lift rate is a real estimator defect (its adaptive gate rejects genuine onsets), flagged as its known weakness.*

Filter	Rest RMS [m/s]	Missed lifts (val)	Censored	Release proxy [s]	Onset proxy [s]	CPU µs
S5_LightGateKF	0.045	0.0%	0	4.23	0.12	18.4
S4_FastSlowGated	0.085	0.0%	0	1.43	0.04	19.0
S1_RobustAdaptiveKF	0.045	17.8–21.6%	2→0†	4.95	0.14	50.7
C03_CV_tracker (ref)	0.048	2.2–2.7%	0	2.18	1.29	7.0
D01_Kalman_CV (ref)	0.096	0.0%	0	2.11	–2.35	16.0

S5 delivers S1-class robustness in constant time (no sliding window) at ~1/3 the CPU; S4 is the fastest exit-from-lift with zero missed lifts. S1P (persistence added to S1) was rejected on validation evidence. Audio-policy tuning on training sessions only (576 policies; mostly lift_on 0.15 / lift_off 0.03 + min_on 0.15) halved the worst false-ON rate (120→60/hr) and transferred to validation without degradation; the safety baseline (max_on 10 s + fault veto) removes the two remaining censored events. With the veto, sustained-fault audio ON-fraction is 0.000 for every candidate (without it: 0.63–0.97) [M][S].

6.3 Synthetic fusion interface and portable C core

The baro/IMU fusion interface (async IMU-predict / baro-correct, gravity compensation with Earth z-down convention a_z = f_z + g, rest-gated accel-bias estimation, baro-only fallback) was verified on seeded synthetic IMU only [S]: climb-rate RMS error 0.499 → 0.009 m/s with known attitude, and 0.034 vs 1.699 m/s during a 1 s baro blackout. Recorded synthetic failures: fusion is slightly worse than baro-only at rest (0.031 vs 0.012 m/s — injected IMU noise has nothing to cancel), and an 8 m baro spike produces 1.50 m/s post-spike velocity. No fusion claim was made for real IMU data until Phase 5.

The portable C99 core (portable/vario_core/, no heap, timestamp-driven, estimator and audio modules independent) reproduces the Python reference on 12/12 test vectors: double build max |Δv| = 2.0e-08 m/s (tolerance 1e-6), float build 5.1e-05 m/s (tolerance 1e-3); it builds with MSVC on host and xtensa-esp-elf-gcc for ESP32-S3 in float and double variants [M/S]. Host runner timing reads 0.000 µs on most vectors (clock() resolution); CPU numbers therefore come from the Python benchmark, and on-target timing remains future work.

7. Phase 5: real-IMU fusion

7.1 Noise decomposition [M]

Direct decomposition of the compensated-altitude noise on the t007 static (holding one ADC channel constant at a time) shows the pressure ADC d1 dominates: 2.56–2.89 Pa lag-1 per sample (0.21–0.24 m), while the clone's temperature channel d2 contributes only ~0.2–0.5 Pa. This refutes the earlier working hypothesis that d2 was a major noise injector (scripts/18_noise_decompose.py). Median-filtering d2 over ±2.5 s before compensation changes total noise by <2 % (kept: physically right, costs nothing). The same decomposition applied to earlier phases' streams: their tuned q/r absorbed this same d1 noise.

7.2 Clone calibration [M]

The MPU6050 clone reads |a| = 10.055 m/s² at rest — a +2.5 % scale error (t008: +2.8 %). The scale factor is a per-device constant taken from the clean static and applied to all sessions; per-session estimates are fragile because slow controlled hand motion keeps |a| inside any plausible band (this poisoned one session's estimate to 9.567). The gyro carries a large but stable offset (≈51.7–55.8 °/s norm, mostly x), removed by capture-start rest-window subtraction.

7.3 Attitude warm start and fault recovery [M]

The 6-DOF Madgwick estimator (β=0.3) is healthy on real data: 0.011 m/s² rest gravity residual after convergence, verified against synthetic controls. But starting from identity quaternion needs ~5 s to acquire the rig's tilt, and the leaked ~2.2 m/s² mean vertical acceleration integrated the fused KF's velocity to 14 m/s; the innovation gate then rejected every measurement for the rest of the session while the fault flag — previously a display-only latch — left the filter coasting at a runaway state. Two fixes in src/: (1) quat_from_accel_tilt, an exact shortest-arc warm start from the first accel sample (az(0) ≈ 0.02 m/s²; roll/pitch trigonometric forms were abandoned after both the plain and conjugated variants left 0.2–0.6 m/s² residual on this X-up rig); (2) loss-of-track reinitialization (8 consecutive >5σ innovations ⇒ reacquire from the measurement, counted in n_reinit).

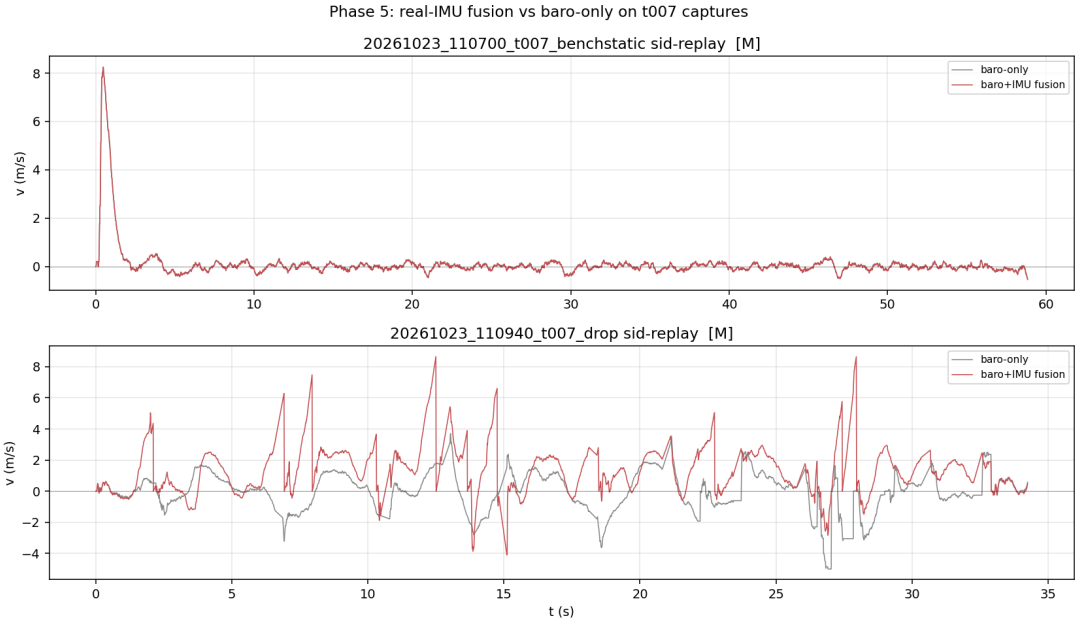


Figure 2. *Phase 5 fused (red) vs baro-only (grey) velocity on the t007 static (top) and elevator (bottom), exploratory q=10 with measured r. At rest the traces coincide; on the elevator the fused track matches baro-only with slightly lower peaks [M].*

7.4 Fusion verdict on real captures [M]

Table 7. *Real-IMU fusion vs baro-only at rest-masked segments, exploratory q=10 [M]. The pattern is monotone in attitude error: fusion is identical at rest, equal on the elevator, worse under hand motion, clearly worse under tumble.*

Session	Baro-only rest v-RMS [m/s]	Fusion rest v-RMS [m/s]	Madgwick |az| [m/s²]	Fusion gates/reinit
t007 benchstatic	0.35–0.76	0.35–0.76	0.011	19–31 / 1–2
t007 elevator_2x	1.33	1.39	0.137	8–12 / 0
t007 ControledMove2	1.14–1.19	1.43–1.47	3.41	329–363 / 13–15
t007 drop / drop2 / toss	1.08–1.64	1.80–3.00	6.5–10.3	308–688 / 13–42

The q sweep on statics is a flat plateau (drift-removed rest v-RMS 0.407–0.447 m/s across q = 0.3…100, spread 10 %) — the metric has no power to discriminate q on office statics — while small q reinit-cycles on motion (q=1: 11 reinit in 34 s on drop). Exploratory operating point q = 10 is a documented judgement call, mid-range and motion-vetted; r = 0.042 m² (t007) is measured [M]. No estimator is selected or frozen.

8. t008 quiet-house static validation

t008_quiet_static (60 s, same rig and firmware 0.7.0 as t007, quiet house, one 1-frame sequence gap) isolates the environment's contribution to the earlier statics. Cross-generation comparison (scripts/21_static_compare.py):

Table 8. *Cross-phase static comparison, both sids [M]. The quiet house removes 7× of slow drift and halves mid-τ Allan deviation vs the t007 office session; the per-sample noise floor is unchanged — it is sensor-intrinsic.*

Capture	OSR / rate	Detrended noise [m]	Drift range [m]	Allan σ(1 s) [m]	Allan σ(10 s) [m]	KF rest v-RMS [m/s]
t005 office	4096 / 45 Hz	0.43	1.3–1.7	0.39–0.40	0.40	0.12–0.13
t006 office	2048 / 90 Hz	0.41	1.2–1.3	0.27–0.29	0.27–0.29	0.10–0.11
t007 office	2048 / 90 Hz	0.44–0.46	9.2–9.8	0.41	0.48–0.51	0.71–0.80
t008 QUIET	2048 / 90 Hz	0.40–0.42	1.3–1.4	0.27–0.30	0.28–0.30	0.10–0.11

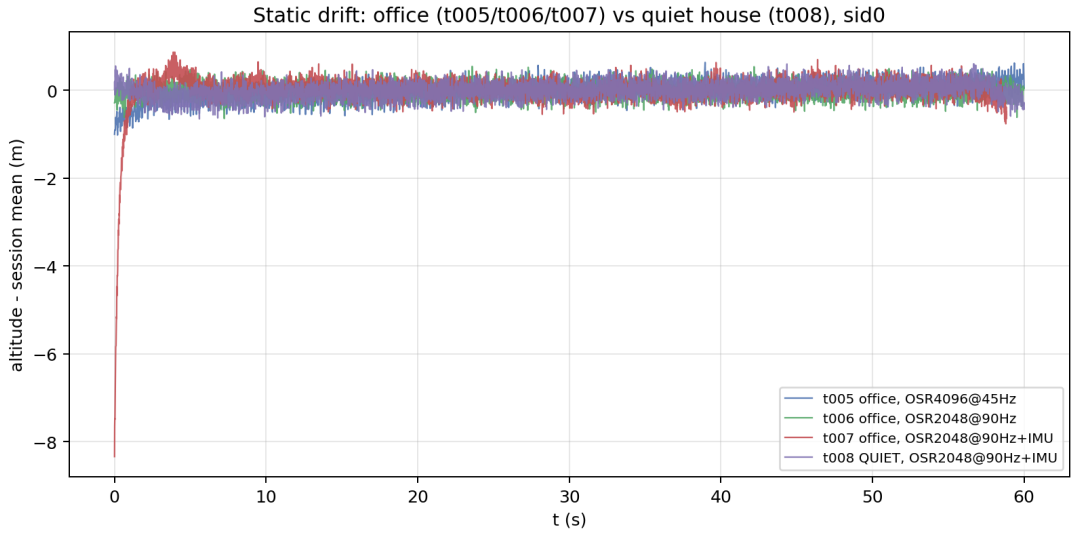


Figure 3. *Altitude relative to session mean for four static captures. The t007 office trace (red) wanders –8.5 m; all others stay within ±1 m [M].*

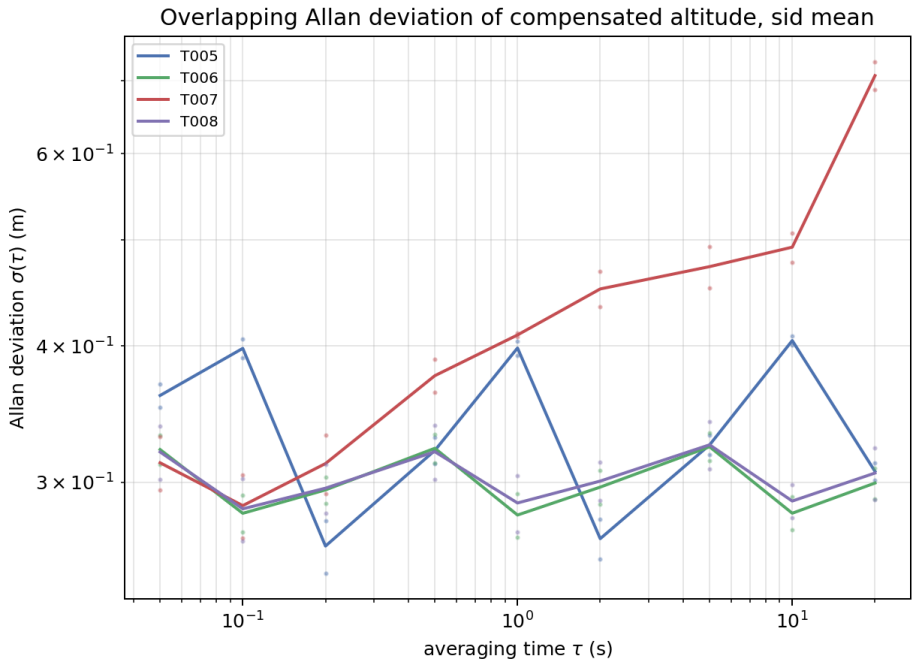


Figure 4. *Overlapping Allan deviation of compensated altitude (sid mean; faint dots: individual sids). All captures share the short-τ white-noise floor; only the t007 office trace rises with τ — random-walk room draft [M].*

Consequences. (1) The KF rest noise floor of 0.10–0.11 m/s is a robust cross-generation measurement of what a quiet MS5611-clone static supports — earlier office numbers conflated sensor noise with draft. (2) The t007-derived r = 0.042 m² was inflated by draft leakage into the 1-s detrend estimator; the quiet-static value is 0.026–0.033 m². (3) Fusion on t008 equals baro-only at rest (0.1085 vs 0.1086 m/s) with zero reinitializations — the Phase-5 machinery reproduces on fresh data. (4) IMU run-to-run consistency between t007 and t008 (scale +2.5/+2.8 %, gyro offset norm 51.7/55.8 °/s, Madgwick residual 0.011 m/s² both) confirms calibration stability across power cycles [M].

9. Failure and correction register

Table 9. *Defects and corrections across phases. Nothing was silently fixed or discarded; frozen defect references keep every failure reproducible.*

#	Defect / correction	Evidence	Status
F1	S2 freeze trap: plausibility pre-clean replaced deviating samples with own prediction → permanent freeze after landings (31 censored beep-offs)	results/11_s2_censoring.csv [M]	Root-caused; saturation fix in S4/S5; S2 frozen as defect ref
F2	Fusion divergence: identity-q0 attitude transient leaked 2.2 m/s² into fused KF → v ran to 14 m/s, gate froze filter; fault flag was display-only	scripts/20_fusion_trace.py [M]	Fixed: shortest-arc warm start + loss-of-track reinit
F3	45-vs-90 Hz rate interpretation in Phase 2 was a session-pair confounder (firmware noise colour), not a rate effect	Phase 3 audit controls [M][S]	Corrected; Phase 2 numbers unchanged
F4	d2 temperature-channel noise hypothesized as major altitude-noise injector	scripts/18_noise_decompose.py [M]	Refuted: d1 pressure ADC dominates (2.6–2.9 Pa vs 0.2–0.5 Pa)
F5	MS5611 compensation scale: a hand-derivation with /2^6 is wrong by ~500×	analysis/common.py [M]	Documented; common.py is the ground truth (final scale /2^15)
F6	Per-session g_cal estimation poisoned by slow hand motion (9.567 m/s²)	script 17 motion rows [M]	Fixed: per-device constant from clean static
F7	S3 audio chatter ~13k ON/hr + 16.2 m/s post-transient; S4/S5 1.3–2.0 m/s excursions on corrupt frames	08_audio_failures.csv, 11_events_corrected.csv [M]	Recorded, unfixed by design
F8	S1 adaptive gate rejects genuine onsets (17.8–21.6 % missed lifts on validation)	results/12_*.csv [M]	Recorded as S1's known defect
F9	USB whole-loop freezes (0.32–0.66 s) in t007 motion sessions	quality report + script 21 gap audit [M]	Documented; MCU pacing work pending; harmless to analysis

10. Limitations, uncertainty, reproducibility

• **No ground truth.** Onset/release/audio timings are derived proxies [D] against smoothed references; negative onset proxies mean a filter anticipated the derived reference, not physical reality. No confidence intervals are claimed; session medians are descriptive, and multiple sids per session are not independent samples.

• **Environmental confounders.** Office statics are draft-dominated (t007: 9 m range); only t008 is pressure-quiet. Cross-capture comparisons therefore report drift-free metrics wherever discrimination matters. Clones vary per unit: the two GY-63s differ ~1.8× in rest noise on some captures, so per-sid evaluation is mandatory and dual-barometer averaging would import the noisier sensor's errors.

• **Motion envelope.** Drop/toss/elevator/controlled-move approximate the flight-relevant envelope partially; no real thermal-lift capture exists yet. Impact saturation at ±8 g limits drop/toss accel fidelity.

• **Fusion limits.** 6-DOF attitude (no magnetometer) degrades under tumble (6.5–10.3 m/s² gravity residual) — the measured fusion ceiling. q=10 is exploratory; the static sweep cannot discriminate q.

Synthetic fusion failures (rest regression, 8 m spike excursion) are recorded.

• **Reproducibility.** Every number traces to a row in results/; figures regenerate from the same CSVs; all capture data and firmware configs are committed; test suite 47/47 (scripts/run_tests.py). Host CPU numbers are relative-cost indicators; on-target ESP32 timing is pending. GitHub pushes from this machine are intermittently blocked by the network; commits are always local-safe first.

Appendix A — Reproduction

Repo: github.com/raghavmalik89/vario-usb-workbook. Captures: analysis/data/*.csv + .json sidecars. Benchmark: filter_benchmark/ (scripts 01–22, results/, reports/, config/, src/, tests/, portable/). Regenerate, in order:

```
python scripts/run_tests.py # 47/47 unit + regression python scripts/02_validate_data.py #
data quality report (22 runs) python -u scripts/03_run_baselines.py # B00-B03 python -u
scripts/04_run_individual_filters.py python -u scripts/05_tune_parameters.py A B C D E
python -u scripts/07_phase3_candidates.py python -u scripts/08_audio_evaluation.py python
-u scripts/09_fault_and_cost.py python -u scripts/11_audio_events_v2.py python -u
scripts/12_estimator_comparison.py python -u scripts/13_audio_policy_tuning.py python -u
scripts/14_imu_fusion_sim.py # synthetic fusion python -u scripts/15_portable_core.py # C
core equivalence python -u scripts/17_real_imu_fusion.py# t007 real-IMU fusion python -u
scripts/18_noise_decompose.py# d1/d2 noise split python -u scripts/21_static_compare.py #
cross-phase statics + t008 python -u scripts/22_t008_fusion.py # t008 fusion check
```

Document generators (this PDF and the companion technical paper): docs/pdf/build_engineering_report.py, docs/pdf/build_technical_paper.py with shared helpers in docs/pdf/build_lib.py. Source reports for each phase: filter_benchmark/reports/phase2_demonstration.md, phase3_engineering_report.md, phase4_final_report.md, phase5_real_fusion.md, data_quality_report.md, dataset_inventory.md; programme handover: docs/HANDOVER.md.